



**KSU Liquid Propulsion
Preliminary Design Review**

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1.1. Document Purpose

This document is designed for the sole purpose of defining and explaining how the liquid propulsion team at Kennesaw State University intends to complete the STRIX engine cold flow test regime. Each subassembly will be defined, explicated, and justified as per the predefined requirements set out by the team. Additionally, all calculations and analysis including the CFD approach will be shown.

1.2. Mission Statement

This mission is designed to support the growth of Kennesaw Aerospace as a whole, introducing a new ability for students to obtain hands-on experience within astronautics beyond their coursework. For the development of the preliminary engine system STRIX – this team will support the development of the students, the team, and the school’s aerospace program. The engine will include an impinging configuration with a 250 lbf thrust. This engine will enable future SRAD motor builds for competitions like IREC, as well as put Kennesaw State within the top undergraduate astronautics' aerospace universities such as Georgia Tech, Embry Riddle, University of Alabama in Huntsville, etc.

1.3 Mission Requirements

The table below defines the overarching requirements for the mission. Each requirement includes its rationale, verification method, with their respective parent and child requirements.

Req #	Requirement	Rationale	Verification Method	Parent	Child
Mission-1	The STRIX program shall complete a fully, successful cold flow and hot-fire test campaign that provides both a competition-ready engine and hands-on team experience.	The first liquid engine in Kennesaw State history ties technical and developmental outcomes to one campaign.	Test	---	SYS-1 SYS-2 STU-1
System-1	The engine shall use an impinging injector	Establishes the baseline	Analysis, Inspection	MSN-1	TST-1

	producing 250 lbf nominal thrust with nitrous oxide (oxidizer) and ethanol (fuel).	configuration that later testing validates.			
System-2	The cold flow test shall replicate hot fire feed pressures and flow rates within an acceptable margin prior to hot fire testing	Validates the feed system in a low-risk environment before hot fire.		MSN-1	TST-1
Test-1	Shall measure injector discharge coefficient (Cd) and verify flow rates are within a defined tolerance of predicted values for both propellant legs.	Confirms orifice sizing and feed system design against the flow model.	Analysis, Test	SYS-1 SYS-2	STU- 1
Student- 1	Give students hands on experience across the design build test lifecycle, including direct involvement in cold flow set up, data acquisition, and analysis	Core student development goals.	Demonstration	MSN-1 TST-1	----

Table 1: Mission Requirements Table

1.4 Summary of STRIX engine development

This engine supports an impinging injector system, featuring nitrous oxide and ethanol as the respective oxidizer and fuel. The engine geometry has been defined as per the engineering drawings shown.

2.0 - Sizing of Injector

The beginning of designing a liquid propulsion rocket engine starts with a concept known as “sizing” the engine which is not to be confused with designing an engine. This has little to do with geometric parameters like the weight of the engine and length of the machine. Sizing the engine is better represented as the abstract chemical process that acts as the starting design. This is the early sandbox of design parameters and input variables that results in a mathematically sound foundation.

2.1 - Fuel Ratio and Input Design Requirements

When selecting fuel and oxidizer, a couple of important factors were considered. The first and most important factor was safety which quickly eliminated Hypergolic Fuel and most cryogenic fuels. The next factor considered was the cost of the fuel and ability to store/handle it. With these requirements staked out, Nitrous oxide and Ethanol qualified as the best candidates. Furthermore, Nitrous oxide has very high vapor pressure which could pressurize the tank with the oxidizer alone. Not having to use a blow-down gas could reduce the complexity of the design and lower the overall cost. Another important property is that Nitrous oxide has a relatively high boiling point compared to other cryogenic fuels at around -88 degrees Celsius. LOX has a boiling point around -180 degrees Celsius, making Nitrous oxide a much easier oxidizer to store and transport. The O/F ratio was determined to be 4.25. Inlet pressure will be 900 psi. The following graph shows how the Characteristic velocity and O/F ratio were optimized using NASA's CEA equilibrium calculator.

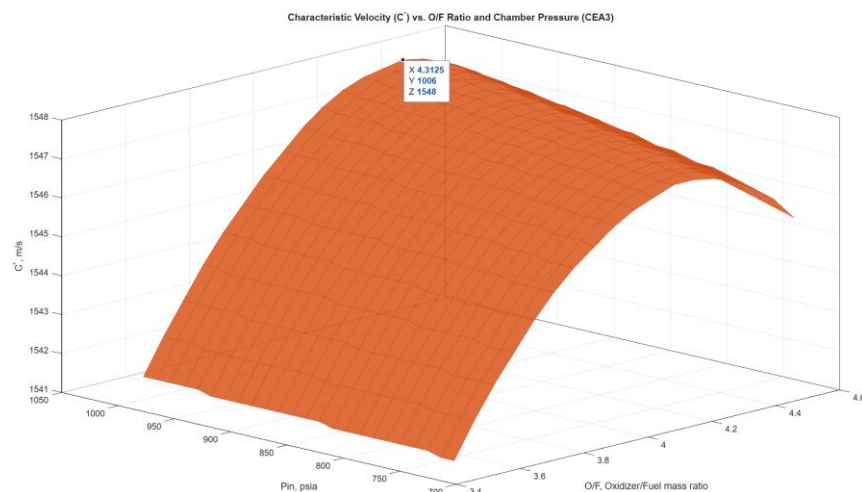


Figure 1: Optimization of O/F Ratio using NASA-GLENN Chemical Equilibrium Program CEA

After fuel/oxidizer was selected, a good starter engine was decided to have around 250lb of thrust. This is because 250lb is a reasonable amount of thrust for our IREC competition rockets. Therefore, once the development of this engine is complete, integrating it into the rocket will be the primary objective. The next step was to decide on Like/unlike-Doublers or triplets. In this case, the unlike triplet had a good mix between high performance and stability because the jets axially impinge at an angle which leads to protection for the chamber walls. The specific configuration was decided as an O-F-O triplet.

Even though some of the input design parameters were not used, these basic inputs were used to establish basic sizing equations. For the goal of a cold-fire test, parameters like propellant state, life requirements, geometric limits, characteristic length, and nozzle configurations were not as important and will be solidified at later stages.

2.1 - Injector Sizing Parameters

The following table covers the basic starting parameters for the sizing the engine. Furthermore, to size the injector plate, the calculations we used to do this are listed below the table.

Parameter	Symbol	Value	Units
Thrust	F	1112.06 (250 lbf)	N
Specific Impulse	I_{sp}	200	s
Gravitational Acceleration	g	9.807	m/s^2
Oxidizer-to-fuel ratio	O/F	4.25	--
Injector Manifold Pressure	P_i	900	psi
Chamber Pressure	$P_f (P_c)$	780	psi
Injector Pressure Drop	ΔP	120 (827,370.88)	Psi (Pa)
Discharge Coefficient	Cd	0.60	--
Ethanol Density (20°C)	ρ_{Et}	789.4 (49.28)	Kg/m^3 (lbm/ft ³)
N2O Density (20°C)	ρ_{N2O}	785.06 (49.01)	Kg/m^3 (lbm/ft ³)

Table 2: Injector Sizing Parameters

Mass Flow Rate Calculations:

- Total Mass Flow Rate ($\dot{m}_{\text{tot}}$)

$$\dot{m}_{\text{tot}} = \frac{F}{I_{\text{sp}} \cdot g} = \frac{1112.06 \text{ N}}{200 \text{ s} \cdot 9.807 \text{ m/s}^2} = 0.566973 \text{ Kg/s}$$

- Oxidizer Mass Flow Rate ($\dot{m}_{\text{N}_2\text{O}}$)

$$\dot{m}_{\text{N}_2\text{O}} = \left(\frac{O/F}{1+O/F} \right) \dot{m}_{\text{tot}} = \left(\frac{4.25}{5.25} \right) (0.566973 \text{ Kg/s}) = 0.458978 \text{ Kg/s}$$

- Fuel Mass Flow Rate ($\dot{m}_{\text{Et}}$)

- $\dot{m}_{\text{Et}} = \left(\frac{1}{1+O/F} \right) \dot{m}_{\text{tot}} = \left(\frac{1}{5.25} \right) (0.566973 \text{ kg/s}) = 0.107995 \text{ kg/s}$

Hydraulics & Orifice Diameter Derivation

- **Governing Equations:** From the continuity equation and incompressible Bernoulli orifice flow:

$$\dot{m} = \rho A v$$

$$v = C_d \sqrt{\frac{2\Delta P}{\rho}}$$

Substitute area $A = \frac{\pi d^2}{4}$ and velocity v into mass flow rate

$$\dot{m} = \rho \left(\frac{\pi d^2}{4} \right) \left(C_d \sqrt{\frac{2\Delta P}{\rho}} \right) = \frac{\pi d^2}{4} C_d \sqrt{2\rho\Delta P}$$

Solving for the single orifice diameter d :

$$d = \sqrt{\frac{4\dot{m}_{\text{orifice}}}{\pi C_d \sqrt{2\rho\Delta P}}} = \sqrt{\frac{4\dot{m}_{\text{total}}}{\pi N C_d \sqrt{2\rho\Delta P}}}$$

- Injection Velocities:

$$v_{\text{N}_2\text{O}} = C_d \sqrt{\frac{2\Delta P}{\rho_{\text{N}_2\text{O}}}} = 0.6 \sqrt{\frac{2(827,370.88 \text{ Pa})}{785.06 \text{ Kg/m}^3}} = 27.55 \text{ m/s}$$

$$v_{\text{Et}} = C_d \sqrt{\frac{2\Delta P}{\rho_{\text{Et}}}} = 0.6 \sqrt{\frac{2(827,370.88 \text{ Pa})}{789.4 \text{ Kg/m}^3}} = 27.37 \text{ m/s}$$

- Calculated Orifice Diameters:

- Oxidizer Orifice(d_{N_2O})

$$d_{N_2O} = \sqrt{\frac{4(0.057373 \text{ Kg/s})}{\pi(0.6)\sqrt{2(785.06 \text{ Kg/m}^3)(827,370.88 \text{ Pa})}}} = 1.8379 \text{ mm}$$

- Fuel Orifice (d_{Ethanol} d_{Et}) $= \sqrt{\frac{4(0.026999 \text{ Kg/s})}{\pi(0.6)\sqrt{2(789.4 \text{ Kg/m}^3)(827,370.88 \text{ Pa})}}} = 1.2591 \text{ mm}$

- **Length-to-Diameter Ratios (L/d):**

- Oxidizer channel length (L/d = 6):

$$L_{ox} = 6 \cdot d_{N_2O} = 8.5(1.8379 \text{ mm}) = 15.62215 \text{ mm}$$

- Fuel channel length (L/d = 3)

$$L_f = 3 \cdot d_{Et} = 3(1.2591 \text{ mm}) = 3.777 \text{ mm}$$

- **Impingement Distance:**

Using the average orifice diameter: $d_{avg} = \frac{1.8379+1.2591}{2} = 1.5485 \text{ mm};$

Impingement distance: $L_{imp} = 5 \cdot d_{avg} = 5(1.5485 \text{ mm}) = 7.7425 \text{ mm}$

- **Impingement Geometry:**

At an impingement half-angle of 45° the transverse orifice spacing from the impingement center is:

$$S = L_{imp} \cdot \tan(45^\circ) = 7.7425 \text{ mm}$$

2.3 Cold Flow Operating Parameters

The table below expresses the parameters for the cold flow test. They are the variables used for the water blow-down test.

Parameter	Symbol	Value	Units
Test Fluid	--	Water (H_2O)	--
Water Density ($20^\circ C$)	ρ_{H_2O}	998.2	Kg/m^3
Manifold Pressure Drop	ΔP	120 (827,371)	Psi (Pa)
Discharge Coefficient	C_d	0.60	--
Oxidizer Orifice Diameter	d_{N_2O}	1.8379	mm

Fuel Orifice Diameter`	d_{Et}	1.2591	mm
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Table 3: Cold Flow Test Parameters

Injection Velocity Cold-Fire Calculations:

$$v_{H_2O} = 0.60 \sqrt{\frac{2(827,371 \text{ Pa})}{998.2 \text{ Kg/m}^3}} = 24.429 \text{ m/s}$$

Mass Flow Rate Calculations:

- For $d = 1.8379$: $\dot{m} = \frac{\pi(1.8379 \cdot 10^{-3})^2}{4} (0.60) \sqrt{2(998.2)(827,371)} = 0.06469 \text{ Kg/s}$
- For $d = 1.2591$: $\dot{m} = \frac{\pi(1.2591 \cdot 10^{-3})^2}{4} (0.60) \sqrt{2(998.2)(827,371)} = 0.03036 \text{ Kg/s}$

$$\dot{m}_{\text{ox circuit, H}_2\text{O}} = 8 \cdot (0.06469 \text{ Kg/s}) = 0.51752 \text{ Kg/s}$$

$$\dot{m}_{\text{fuel circuit, H}_2\text{O}} = 4 \cdot (0.03036 \text{ Kg/s}) = 0.12144 \text{ Kg/s}$$

$$\dot{m}_{\text{total, H}_2\text{O}} = 0.51752 + 0.12144 = 0.63896 \text{ Kg/s}$$

It is important to note that $C_d = 0.6$ is an assumption for a rough wall, and the true tested value will be compared alongside CFD results to calculate true manufactured C_d .

3.0 - Injector CAD

The next step in designing the injector is to take the defining equations for sizing and create CAD models to successfully communicate the design for manufactures.

3.1 - Injector Design Choices

After considering injector outlet sizing parameters, the team designed a triple unlike impinging injector manifold while keeping designing within the constraints of two oxidizer manifolds and one fuel manifold, ¼ NPT inlets, material capable of sustaining pressures

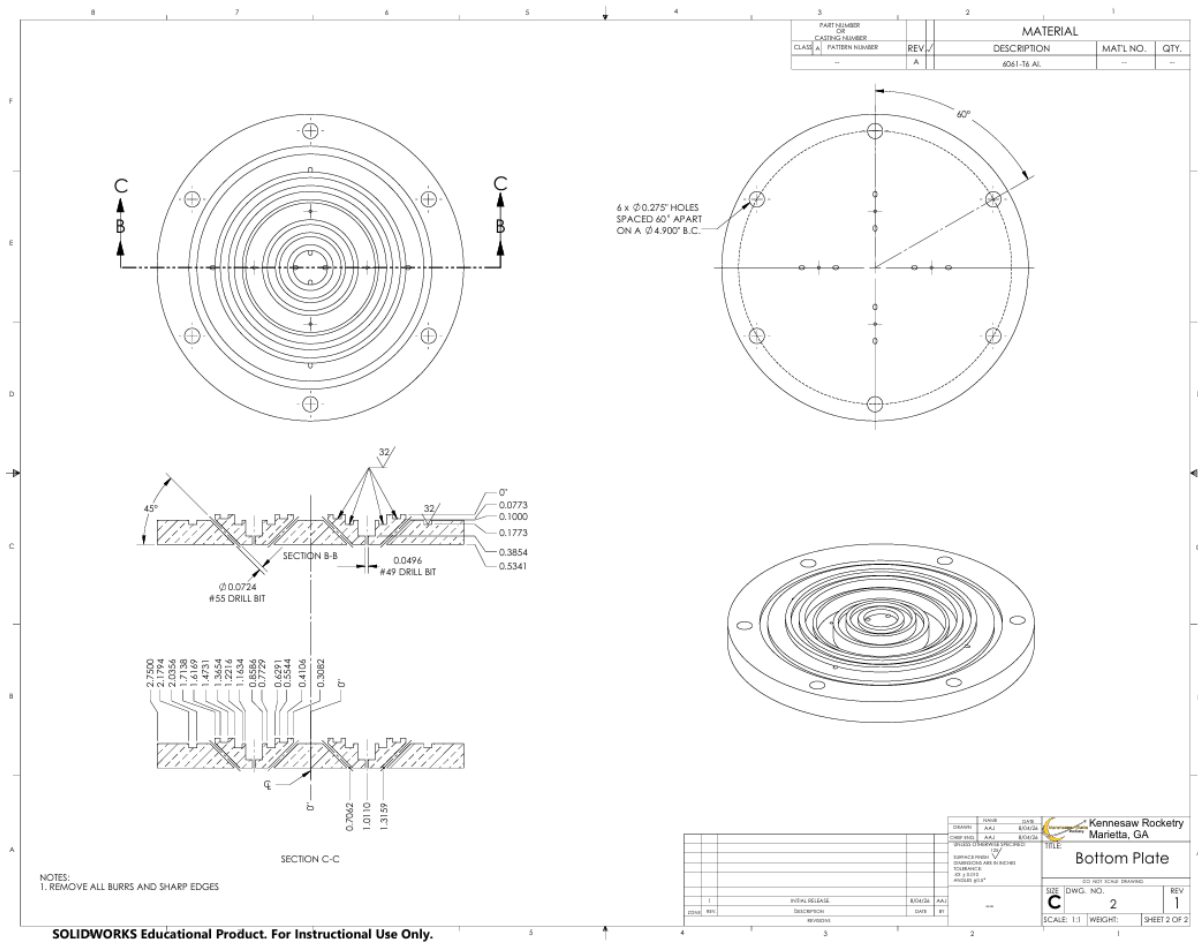


Figure 3: Liquid Propulsion Bottom Plate

4.0 - Plumbing Decisions

After creating the CAD for the Injector Plate, the next step in designing a liquid propellant engine is to create a plumbing system to get fuel and oxidizer to the injector.

4.1 - Feed system

When choosing the feed system, safety and simplicity were the most important considerations. This immediately eliminated a pump fed system due to the high pressures, complexity, and expensive parts. Most pump-fed engine cycles rely on using fuel that was re-routed out of a regenerative jacket, or exhaust from the inside walls of the combustion chamber to spin up the pump turbines. This type of pump is referred to as a Tap-off cycle, and it uses bled-off combustion gasses from the combustion chamber to spin up the fuel

pump. There are many other similar feed systems that have high performance and efficiency benefits. However, for the scope of KSU's first liquid propulsion engine, a blow-down system was selected.

4.2 - Testing Purposes & Plumbing

The initial testing stage of project STRIX has been determined to test and observe spray and droplet dynamics within our modeled injector. To do this, the team has decided to conduct a cold flow test involving a 120 psi drop with water at room temperature. Reasoning was determined to be due to the water's similar density with both N2O and ethanol. The test will also determine atomization characteristics within the combustion chamber by determining SMD (Sauter-Mean diameter), velocity profile, spray direction, and impinging point predictions through basic calculations and CFD results. These results can be deliberated further through pressure transducer information during any specified timestamp. Electronics and initial plumbing charts are shown below:

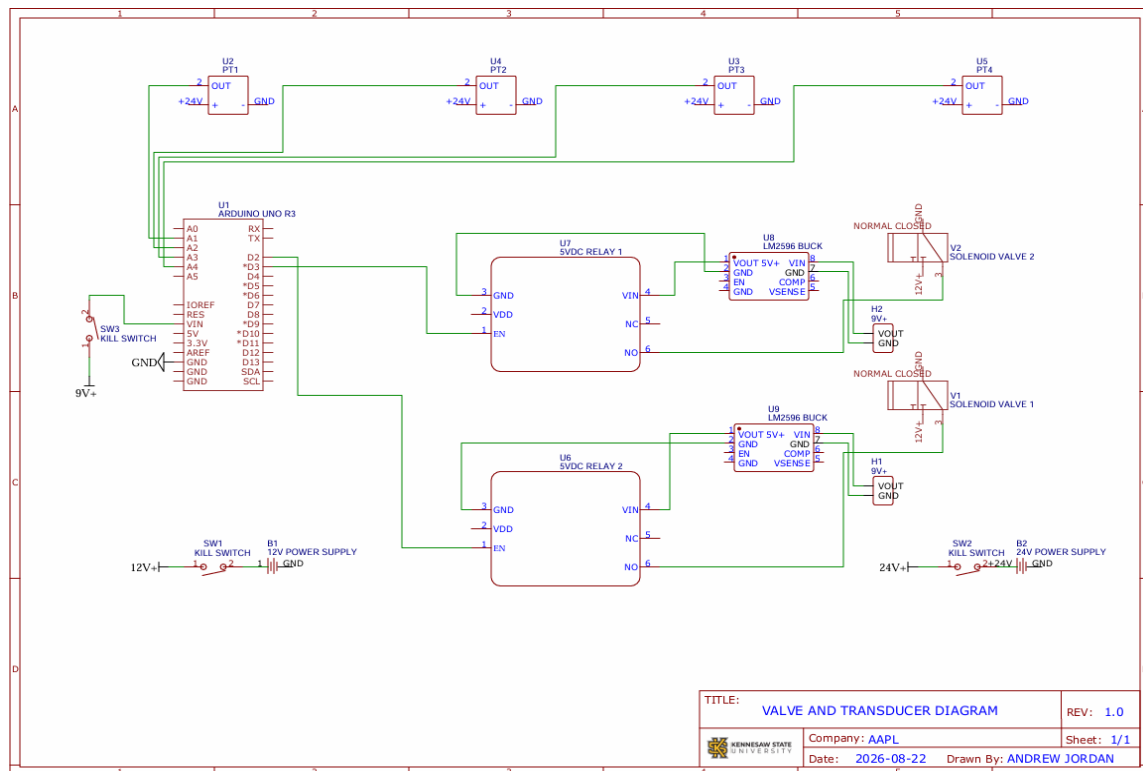


Figure 4: Valve & Transducer Diagram

The plumbing system features solenoid valves which route from an air compressor to a water tank, which then diverges into three separate lines for an initial inlet pressure of 134 psi. Combustion chamber pressure is assumed to be 1 atm, or approximately 14 psi. The plumbing resembles a blow down system, where pressures may fluctuate upon opening the solenoid valve. This large breadth of pressures will give variations of velocity, SMD, and potential spray direction characteristics that will be investigated for use on a static fire procedure. Reynolds numbers will be reported alongside pressure.

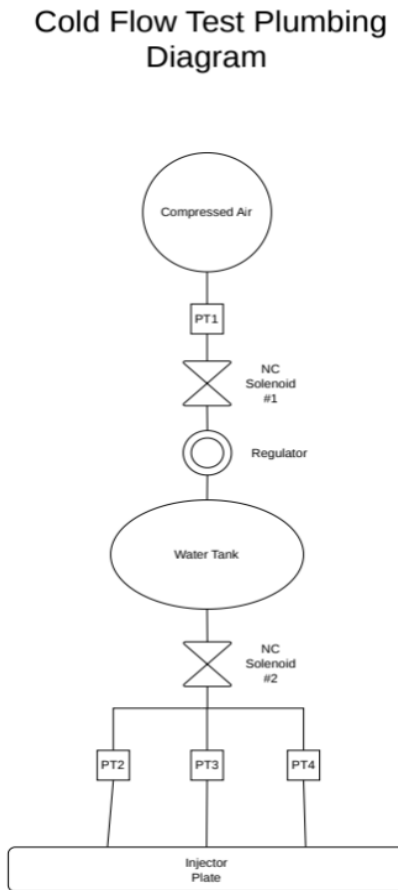


Figure 5: Plumbing Diagram

4.3 - O-ring calculations

Injector geometry is also optimized for O-Ring to squeeze and fill percentages. Through use of the following calculations, the O-Rings determined for use are as follows:

GOVERNING EQUATIONS

Quantity	Formula	Notes
Gland fill, F%	$F\% = (A_0 / G \cdot W) \times 100$	A_0 = O-ring cross-section area, G = gland depth, W = gland width. 70–80% optimal per Parker ORD 5700, Sec. 4
Squeeze value, S_v	$S_v = d_2 - G$	d_2 = O-ring cross-sectional diameter (CS)
Squeeze percent, S%	$S\% = (S_v / d_2) \times 100$	15–30% optimal, target ~25%
Cross-section area	$A_0 = \pi d_2^2 / 4$	Used in fill% above

Table 4: O-Ring Equations

DESIGN VALUES

d_2 (CS)	G (depth)	A_0	$S_v / S\%$	F% at nominal W
0.103"	0.07725"	0.00833 in ²	0.0258" / 25%	75%

CONCENTRIC BORE DIAMETERS

D1	0.965
D2	1.402
D3	2.587
D4	3.040
D5	3.965

Table 5: Resultant Bore Diameters

O-RING / GROOVE SCHEDULE

Ring	AS568 Dash	ID (in)	CS (in)	OD (in)
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R1	AS568-118	0.862	0.103	1.068
R2	AS568-125	1.299	0.103	1.505
R3	AS568-145	2.550	0.103	2.756
R4	AS568-151	2.987	0.103	3.193
R5	AS568-166	6.737	0.103	6.943

Table 6: Resultant O-Ring Parameters

5.0 - Cold Fire Testing and set-up

5.1 - Facility and Equipment requirements

An air compressor suitable for the task as well as access to power and water. If possible, an area that is clear from people for safety

5.2 - Test Stand

The test stand will consist of a Home Depot™ plywood backer board to which the reserve tank, injector, and hoses will be fixed. Simple air hoses can be used to connect the compressor to the reserve tank; larger diameter hoses may be needed to connect to the injector's T fitting. A pattern may be used behind the injector flow for contrast and velocity calculation. The spray pattern will be recorded with a high-speed camera.

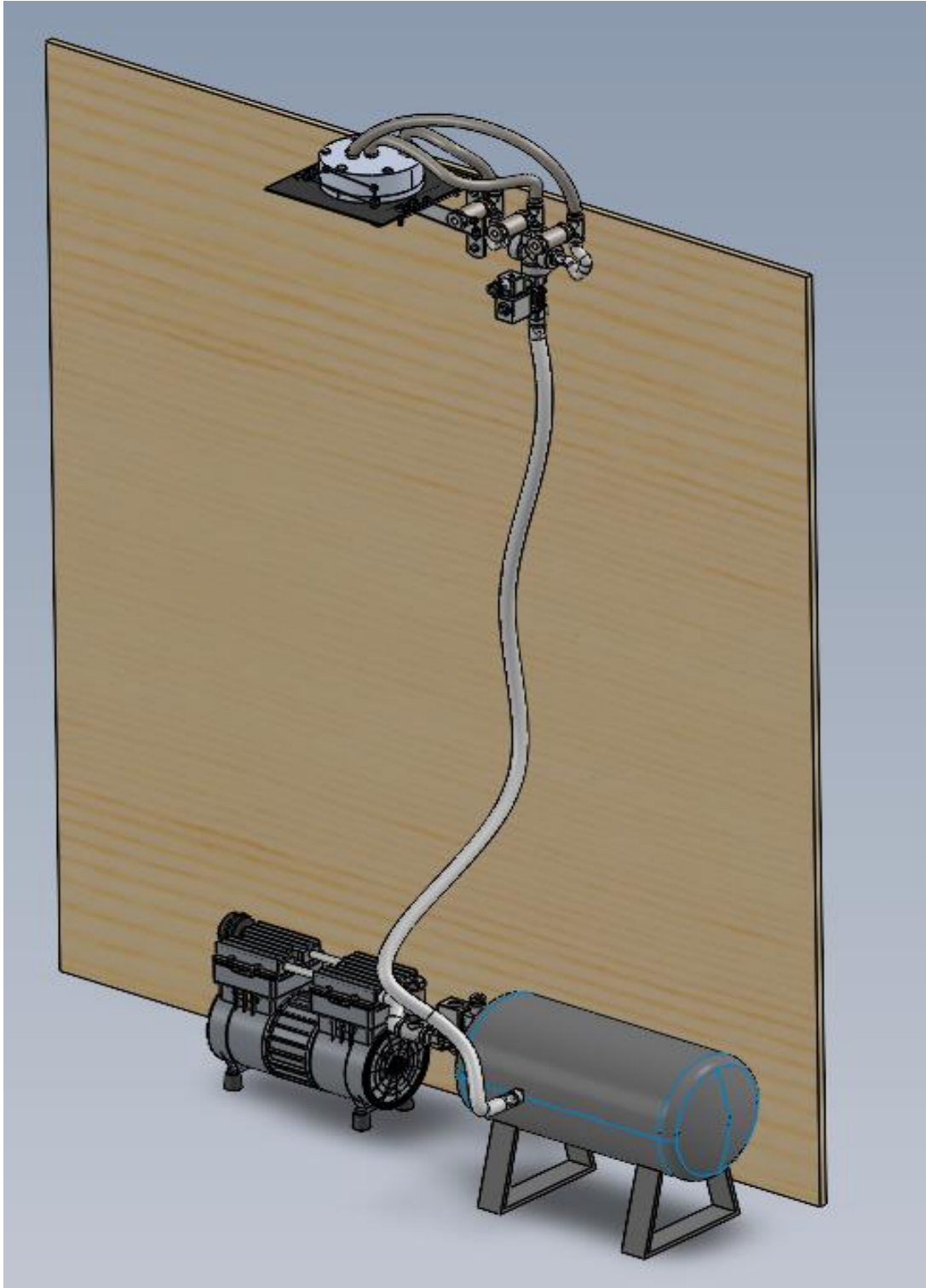


Figure 6: Test CAD Render

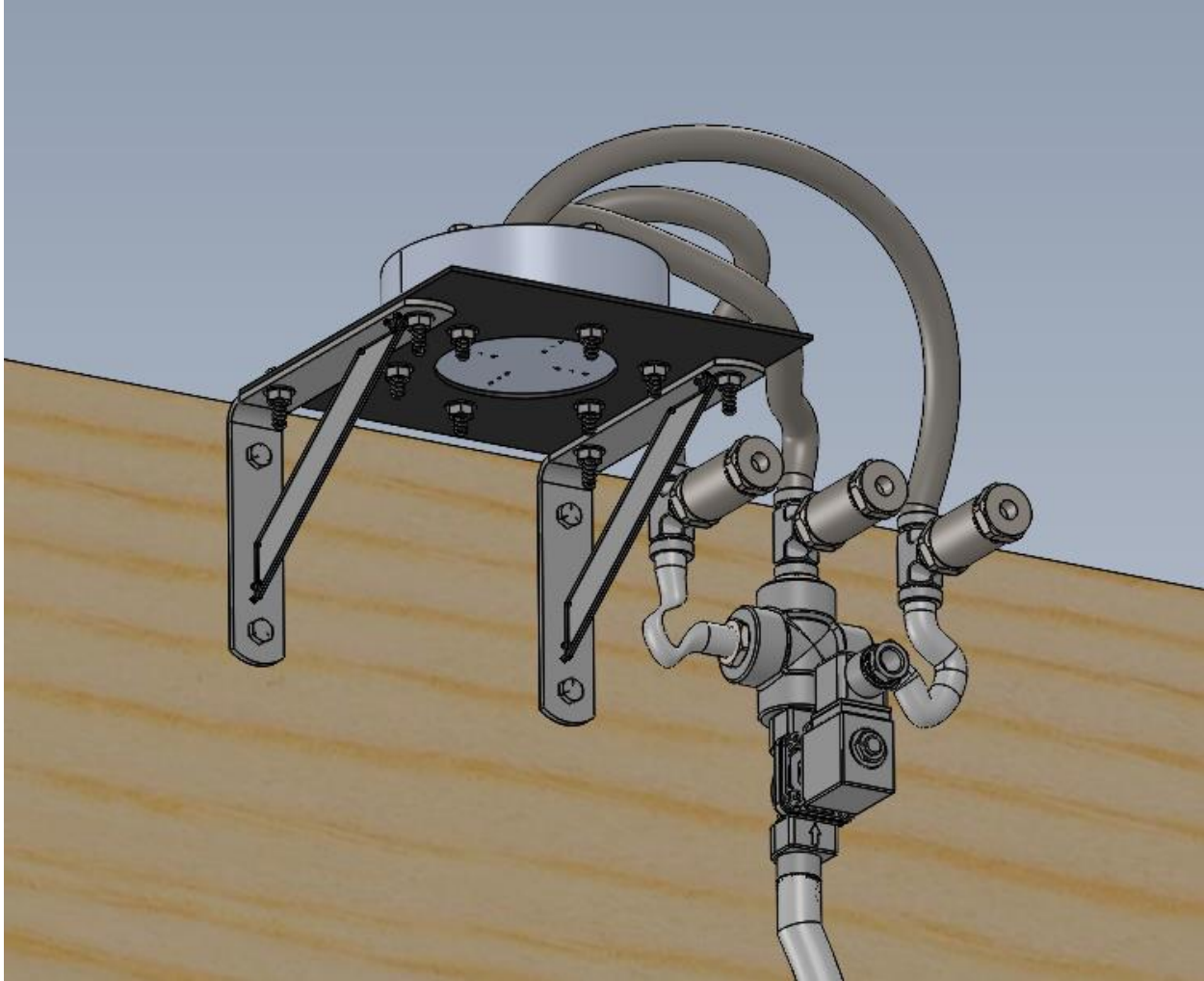


Figure 7: Downward Facing Configuration

5.3 - Test desired results

As this is a test and not for the purpose of reinforcing known information, there are only a few values that are desired to be attained with known previous information – a velocity profile approaching 24.429 m/s, and SMD values that are within 10-20% of CFD reported values. Actual SMD values are expected to be within the ballpark of 85-90 μm , however this value is another data point which will be observed during testing and CFD. Additional results will be compared and desired as CFD completes, particularly Reynolds number and the resulting velocity and spray profile with this value.

The primary reasoning behind the test is to verify that the injector is manufactured properly and is within an acceptable margin of error with CFD analysis to proceed with a static fire test.

6.0 - Cost Analysis

In the section, the cost of the cold flow test will be discussed. There will be data tables and descriptions on what needs to be purchased and how much the total estimate cost will be.

6.1 - Cost Analysis Chart & Reasoning

The following chart displays the item and associated cost. Further deliberation on each item is shown:

Item	Qty	Est. unit	Est. subtotal
Air Compressor (already own)	1	—	\$0
½ NPT brass solenoid on/off valve	2	\$36.39	\$72.78
Arduino (already own)	1	—	\$0
Pressure transducer	4	\$31.99	\$127.96
¼ NPT Nipple	3	\$5.69	\$17.07
½-¼-½ tee	1	\$10.12	\$10.12
¼ Tee	4	\$7.76	\$31.04
12V power supply	1	\$19.90	\$19.90
24V power supply	1	\$39.99	\$39.99
Relay	1	\$7.99	\$7.99
Buck converter	1	\$7.99	\$7.99
9V batteries	1	\$12.69	\$12.69
Kill switch (24V & 12V)	2	\$16.99	\$33
Braided Steel Hose	2	\$30	\$60
Top Plate 6061-T6 Stock	1	\$58.35	\$58.35
Bottom Plate 6061-T6 Stock	1	\$40.59	\$40.59
¼" Bolts	1	\$9.48	\$9.48

<u>Male-to-Male Port ½ NPT</u>	1	\$10.49	\$10.49
<u>¼ to ½ NPT</u>	1	\$15	\$15
<u>4-Way ½ NPT Port</u>	1	\$47.59	\$47.59
<u>¼ NPT Hose</u>	5	\$32.99	\$164.95
Subtotal			\$786.98

Table 7: Cost Analysis Table

An important note to share regarding additional costs includes possible manufacturing costs. Protospace MFG is a potential supplier; however, online quotes are far from accurate. Noting the team's \$1,000 credit from IREC, the most difficult part to machine, namely the bottom plate, may be outsourced to this manufacturer – with the team manufacturing the top plate in house. Both stocks are included as well to supply the team with potential to directly contact machine shops within the Marietta area in the event of not being able to access in-school machines. Currently the team plans to machine the plates in house and will keep the Protospace credit in mind for the worst-case scenario.

Each part is included and required with use in previous diagrams for both plumbing and electrical use. High speed cameras and other equipment are not included as these items will be rented from a third-party source. As the team plans on presenting at the AIAA Student Conference II, which requires an abstract submission by late January, this is not an immediate issue – though something to keep in mind as manufacturing takes precedence.